Photo-doping of graphene enhanced by stable perovskite and hole transfer layer

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Recently, researchers in many fields have focused their study on graphene as a potential candidate material for the channel of the field effect transistors (FET) due to its extraordinary carrier mobility [1]. Furthermore, graphene is a popular plasmonic material due to its unique electrodynamic response [2].

Particularly, it has been suggested to use graphene-based FET (GFET) devices as a photo-transistors for photo-sensing applications [3]. However, despite of the extraordinary mobility of carriers in graphene, the responsivity of the devices – the most important characteristic of photo-detectors – is lower than that of the III-V compound semiconductor-based photo-detectors, stemming from the fundamentally limited light absorption efficiency in graphene of just 2.3%. Some researchers suggest to improve the light absorption efficiency in GFETs by introducing quantum dots, perovskite, and other sensitizers, demonstrating significantly improved responsivity due to the photo-doping effect in graphene [4]. At the same time, the conductance between the graphene and the sensitizer layer may be reduced because of the difference in LUMO (or HOMO) energy levels between the materials. Furthermore, while conventional spin-coating deposition of perovskite is usually performed on hydrophilic surfaces, graphene has low surface polar energy and is hydrophobic [5]. Therefore, further improvement of both GFET structure and fabrication technique is of great necessity.

In this study, we introduce additional layers of MoO_3, PEDOT:PSS, and $MAPbI_3$ into the GFET structure in order to improve hydrophilicity, band energy alignment, and light absorption efficiency respectively. Our devices demonstrate improved normalized responsivity as compared to previous works. In addition, we consider an increment of the photocurrent over the dark current value to quantify the photo-doping efficiency in saturated photocurrent regime, which shows a 20% increase of photo-doping efficiency within hole-doped regime comparing to previously reported results. Our results are important for further development of photo-doped graphene-based devices operating in saturated photocurrent regime, e.g. plasmonic gratings, metasurfaces, modulators, etc., which operation is directly controlled by doping.

References

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